Received: 30 September 2016 | Revised: 31 January 2017 | Accepted: 8 February 2017

DOI: 10.1111/coin.12122

WILEY Computational Intelligence

ORIGINAL ARTICLE

One-class SVM for biometric authentication by keystroke dynamics for remote evaluation

Thierry Eude | **Chuan Chang**

Department of Computer Science and Software Engineering, Laval University, Québec, Canada

Correspondence
Thierry Eude, Department of Computer Science and Software Engineering, Laval University, Québec, Québec, G1V 0A6, Canada.
E-mail: thierry.eude@ift.ulaval.ca

Funding information
Cisco

Abstract

Remote skills assessment in distance education needs individual identification in distinguishing between candidates and impostors. Keystroke dynamics is a behavioral biometrics which can be used to identify them. To expect lower error rate, behaviors should be as natural and consistent as possible. The unique identifier assigned to students at their registration seems appropriate but the classification method applied for this case of anomaly detection must be robust even with a lower signature number. In this paper, we first explain how we construct our own dataset. Three methods of selecting Gaussian kernel parameters for one-class support vector machine are subsequently studied regarding the targeted application constraints. The results show that an indirect method as distance to farthest neighbor cannot be used because some signature features have multimodal and dispersed distributions. A method is then proposed based on the selection of the parameters via detecting the "tightness" of the decision boundaries and uses a greedy search. Its performances are compared to those of a grid search method using LibSVM. The results show that the proposed method is more robust when the signatures number decreases and better and more stable in detecting impostors.

KEYWORDS

Gaussian kernel hyper-parameters selection, keystroke dynamics, one-class classification, remote evaluation, support vector machine

1 | INTRODUCTION

At present, 1 major problem for distance education is skill assessment. To give credit to students, it is indispensable to ensure that the conditions are equivalent to the bricks-and-mortar version.

 © 2017 Wiley Periodicals, Inc.

To guarantee learning results, special attention must be devoted to proctoring and the problem of cheating. This needs a specific process, including authentication, oversight of impostors who could substitute for the student, and supervision of unauthorized access of documents. This article focuses on the authentication.

The authentication consists in determining whether someone or something is who or what it claims to be.[1] Such process can fail in 1 of 2 ways: either an authorized user is rejected or an illegal user is incorrectly granted access to the system. Biometric authentication is an automatic method of verifying the identity of an individual based on the measurement of his or her unique physiological traits or behavioral characteristics.[2] Because biometrics cannot be lost, stolen, or overheard like keys or passwords, it is an excellent candidate for identity verification.[3]

Biometrics can be generally divided into 2 categories: physiological biometrics, which are biological/chemical traits that are innate or naturally developed (face, palms, iris, etc) and behavioral biometrics, which are mannerisms or traits that are learned or acquired (voice, handwriting, signatures, keystroke dynamics, etc).[2]

Keystroke dynamics is a behavioral biometrics based on the manner and rhythm in which each individual types.[2] It possesses multiple advantages: uniqueness, low implementation and deployment costs, transparency, and non-invasiveness.[4] It can be used to identify students for remote evaluations. To expect lower error rate, behaviors should be as natural and consistent as possible. But the classification method used for this case of anomaly detection must be robust even with a lower signature number. The organization of this paper is as follows. The next section introduces the process steps used for individual identification. After the explanation of the data collection in Section 3, Section 4 presents the proposed methodology. The results are shown in Section 5, and Section 6 concludes the paper.

2 | KEYSTROKE DYNAMICS IDENTIFICATION

2.1 | Input data

Keystroke dynamics can be applied as either static or dynamic text input. The first one requires the user to enter a predefined text string. The other allows the user to enter any text in an unconstrained way and involves continuous or periodic monitoring of his or her keystroke behavior. During the last 3 decades, the state of the art of keystroke dynamics says that static text input provides much more acceptable error rates and is significantly easier to be implemented.[5] Robinson et al have obtained good results using only first names of an average length of 6.4 characters.[6] Bleha et al used users' first and last names to get slightly better results. In this case, the string length doubled (the length of a full name varies from 11 to 17 characters).[5]

Otherwise, to expect lower error rate, behaviors should be as natural and consistent as possible. Indeed, asking to enter a text that the user does not usually do increases variability. Then, common data entry collected during different times of the day and different times of the week reduces the influence of stress that might imply identification at the beginning at an early stage of assessment.

Regarding the case of Laval University, the use of the IDUL* seems appropriate. All members of Laval University get a unique identifier (IDUL) at their registration. The IDUL is composed of 5 letters which were generated from their first name and last name, followed or not by up to 3 digits. The IDUL is not confidential information, while the password created by the member should always

*IDentifiant Universite Laval (IDUL).

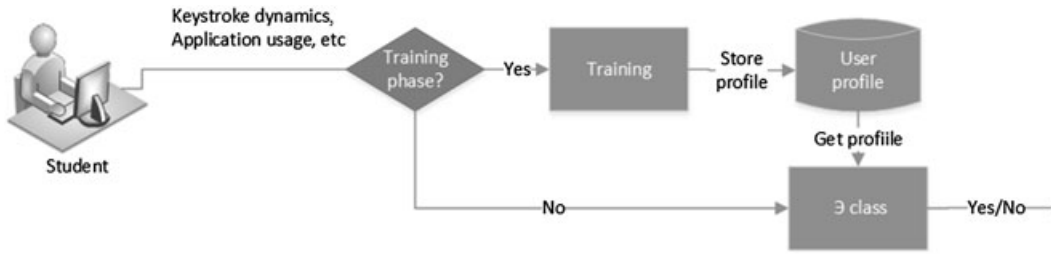


FIGURE 1 The 2 phases for authentication process by keystroke dynamics (training phase and test phase) [Color figure can be viewed at wileyonlinelibrary.com]

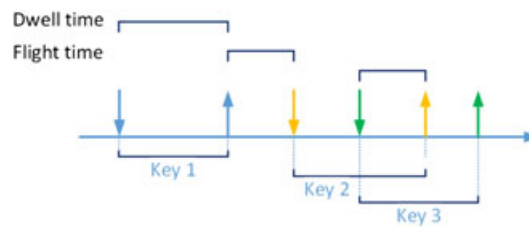


FIGURE 2 Feature vectors in keystroke dynamics: the down and up arrows represent respectively the instants of pressing and releasing of each key [Color figure can be viewed at wileyonlinelibrary.com]

be kept secret. The IDUL and the password together give the member access to his or her personal data and to necessary computer systems for the realization of online activities at Laval University (eg, online self-service system for studies, email, online library, the course portal, etc). To provide an identification solution that integrates easily with the existing system, we chose the IDUL to identify the user, which is a very common static string for each member and can be entered regularly and naturally.

Furthermore, as shown in Figure 1, the authentication process by keystroke dynamics can be logically divided into 2 phases: enrollment phase and verification phase, respectively called training phase and test phase in the domain of classification. During the enrollment/training phase, user's biometric data is acquired, processed, and stored as a pattern in a database for future use by the system. During the subsequent verification/test phase, new user biometric data is acquired and processed. Therefore, the authentication decision will be based on the output of the matching process of the newly presented biometric with the pre-stored reference patterns.[2,7]

2.2 | Feature extraction

The instants when each key is pressed and released are stored as basic data of keystroke dynamics. Two feature vectors based on the data are then generated as input for the classification algorithm (Figure 2):

- Dwell time: interval between the instants when a key is pressed and released. It represents the time that the key keeps being pressed.
- Flight time: interval between the instants when a key is released and the next is pressed. Sometimes it can be negative.

A high resolution time API in JavaScript was used to improve the time resolution up to 1 ms. Normally, higher resolution of the captured data results in better classification accuracy.

2.3 | Classification with support vector machine

A classification task usually involves separating data into training and testing sets. Each instance in the training set contains 1 target value (ie, the class labels) and several attributes (ie, features of the keystroke duration).

An SVM† constructs a hyperplane in high or infinite dimensional space for classification.[8] The goal of SVM is to produce a model based on the training data which predicts the target values of the test data, given only the test data attributes. It is normally used for binary-class classification. For example, given training set for instance-label pairs $(x_i, y_i), i = 1, \ldots, l$, where $x_i \in R^n$ and $y \in \{1, -1\}^l$, the SVM requires the solution of the following optimization problem:

$$\begin{aligned} &\min_{W,b,\xi} \frac{1}{2} W^T W + C \sum\nolimits_{i=1}^{l} \xi_i \\ \text{subject to} \quad & y_i(W^T \text{Œ}(x_i) + b) \geqslant 1 - \xi_i, \\ & \xi_i \geqslant 0. \end{aligned} \tag{1}$$

Thus, each vector x_i is mapped into a higher or even infinite dimensional space by the function ϕ. SVM finds a linear separating hyperplane with the maximal margin in this higher dimensional space. $C > 0$ is the penalty parameter of the error term. Furthermore, $K(x_i, x_j) \equiv \phi(x_i)^T \phi(x_j)$ is called the kernel function. With the kernel trick, the computational efficiency has been greatly improved and the SVM has been extremely successful as a nonlinear classifier. Four basic kernels often used are linear, polynomial, radial basis function (RBF), and sigmoid.[9]

Then we search for an appropriate kernel function and the best values of hyper-parameters to achieve a good separation by the hyperplane in feature space that has the largest distance to the nearest training data point of any class. As a result, the separating function is able to create complex boundaries in original space. Thanks to its superior performance in handling large dimensional data without increasing the system complexity, the SVM has been extensively used in a wide variety of applications. For example, it was adopted for distinguishing impostor patterns by creating a margin that separates normal patterns from impostors by many studies in the state of the art of keystroke dynamics. Support vector machine has a competitive advantage regarding performance compared to neural network.[4] The LibSVM[10] is apparently a good and simple tool available for classification for keystroke dynamics, as an open-source and widely used machine learning library.

3 | DATA COLLECTION

Because of the particularity of IDUL and the difference between each IDUL, we could not reuse any existing dataset.

3.1 | Positive data

Instead, we constructed our own dataset of IDULs with the students who took 1 remote course in the same class at Laval University.

At this stage, the collection of keystroke dynamics data will be done at every access to educational materials. At each connection for visiting the page, we ask the student to input his or her IDUL to build his or her keystroke signature. In this way, the collection of signatures is not affected by the ripple

†support vector machine.

effect that appears when the same character string is input several times consecutively. The strings (IDUL) and their temporal features of keystrokes (dwell time and flight time) will be calculated and stored in secure data files to which only the researchers have access. A validation process is used to ensure the quality of input data. Afterward, the student will be asked if he/she accepts or not to store his/her data for future use in other works. By clicking on a "yes" or "no" button, the response will be stored. Then it will automatically redirect to the page of teaching materials. However, at the first connection of the page, a student could click the link "Do not participate in this research" to refuse the participation.

Since all keystroke signatures were input by the students on their own, this keystroke dynamics dataset is called positive data in this research.

3.2 | Negative data

To test the performance of the SVM classifier (measured by the rate of successful classification of positive and negative signatures), we must collect negative data which is in fact false students' keystroke signatures. These false signatures simulate the inputs that would come from impostors trying to log in with students' IDULs.

To get as close as possible to the collection condition of positive data, we perform the collection of negative data on several sessions for several days, avoiding training users on the keystroke signatures. We first asked 20 people to create our negative data. They had to type the IDULs of the students who participated earlier in the positive data collection at the first stage of the study. When it was possible, we searched participants with similar characteristics (age or keystroke speed) to our initial students (positive data) to have representative data.

To collect all of this data, we created and implemented webpages where the participants could enter the false keystrokes signatures. The participants were asked to access the webpage where IDULs were listed. Then they had to retype these different IDULs coming from our dataset of positive data. The IDULs and their temporal features of keystrokes (dwell time and flight time) were extracted and stored in secure data files as was done for the positive data collection.

In total, we had 3 different webpages with 20 IDULs that had to be retyped each time. The participant was given the indication to access the 3 webpages at 3 different moments during the day or on different days. We asked them to do so as a way to repeat the action of the original students who connected to the log in a page at different times during the day. Assuming that the moment during the day could affect the behavior of a student's typing, we wanted to collect the negative data that would be as representative as possible.

4 | METHODOLOGY

4.1 | One-class classification

The target application is the identification of a student wishing to take an exam. It aims to determine, from their keystroke, if he/she is the right student or an impostor who wants to take the exam. In fact, it is not always possible in practical conditions to obtain sufficient negative data created by impostors for building models. Thus, according to the targeted application, it relates solving a one-class classification problem. As a matter of fact, for one-class SVM (OCSVM), it is unnecessary for its training set to contain data patterns of impostors. The negative keystroke signatures were only collected for testing the performance of SVM classifier. After learning the authorized user's data patterns, OCSVM can find the domain where his patterns probably live and impostors can be detected as outliers because their patterns are outside the domain, which is also called novelty detection.

Solving the OCSVM optimization problem is equivalent to solving the dual quadratic programming problem:

$$\min_{\alpha} \frac{1}{2} \sum_{ij} \alpha_i \alpha_j K(x_i, x_j) \tag{2}$$

subject to the constraints $0 \leqslant \alpha_i \leqslant \frac{1}{\nu l}$ and

$$\sum_i \alpha_i = 1,$$

where α_i is a Lagrange multiplier, ν is a parameter that controls the trade-off between maximizing the distance of the hyperplane from the origin and the number of data points contained by it, l is the number of points in the training dataset.

4.2 | RBF kernel and tuning of hyper-parameters

As mentioned before, the OCSVM first requires the choice of a kernel and certain hyper-parameters to define a frontier. Since the distributions of both dwell time and flight time of keystrokes are continuous, the RBF kernel is in general a reasonably good choice to deal with nonlinear and multimode distributions, [8,11] with

$$K(x_i, x_j) = exp(-\gamma \|x_i - x_j\|^2), \gamma > 0, \tag{3}$$

where γ is the only hyper-parameter of RBF kernel, which makes it simpler than other kernels.

The hyper-parameter γ of RBF kernel plays a key role in classifiers. The effect of γ on the tightness of decision boundaries is monotonic. With increasing of γ, the RBF kernel will be narrower, the number of support vectors increases, and the decision boundaries become tighter. Meanwhile, the false negative rate increases and the false positive rate decreases. If γ is very large, all training vectors will end up as support vectors, which is called overfitting and is clearly not desirable. Otherwise if γ is very small, the frontier will be too loose and underfitting appears. Thus, we could get a weak accuracy in the training phase, and it will also decrease the accuracy in the testing phase.

On the other hand, the parameter $\nu \in [0, 1]$ is used to tune the upper bound on the fraction of outliers outside the decision boundaries and the lower bound on the fraction of support vectors in the training dataset.[12] Therefore, it is still possible for OCSVM to contain outliers in the training data, because the training data may be polluted by noises caused by the measurement.

4.3 | Experimental protocol

To reduce the noises existing in the measurement of the dataset, a preprocessing of data is necessary. According to Joyce and Gupta,[13] all keystroke timing data greater than 3 standard deviations (σ) above the mean reference signature are discarded because they denote an unusual behavior proper to the individual.

Then, as shown in Figure 3, we divide the positive data into 2 parts (4/5 + 1/5). The signatures in the second part (1/5) are selected randomly from the whole positive data. The first part (4/5, called Data 1) is used for training the model in OCSVM. The second part is mixed with the negative data (together called Data 2) for testing the obtained classifier model in binary-class SVM. Here, we use as many negative signatures as positive ones for each IDUL. To avoid the influence of random selection of data to the classification process, we repeat the positive data separation to further conduct the experiment 3 times.

During the training phase, we use Data 1 and RBF kernel to train the classifier model for each IDUL. The best value pair of hyper-parameters (γ, ν) is searched to find the best model. In this phase,

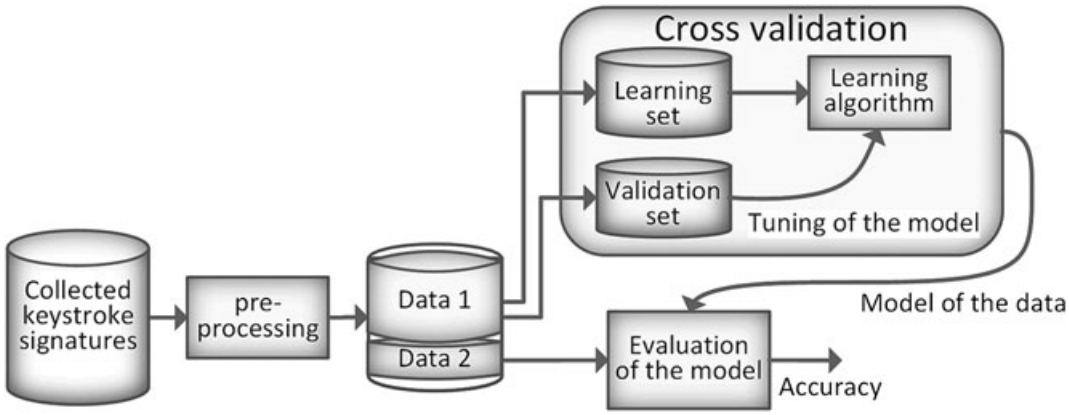


FIGURE 3 Learning and model evaluation process

a 5-fold cross-validation[‡] is applied to relieve the overfitting problem. Then, a grid search method is used to search for the best value pair (γ, ν) for each IDUL with LibSVM. This could be regarded as tuning the frontier of classification by trying out all combinations of γ and ν values in a certain range to see the results. Finally, we chose the value pair (γ, ν) with the best performance measurement achieved. It is worthwhile to note that the performance measurement of LibSVM "accuracy[§]" becomes "recall[¶]" in our one-class training phase.

During the test phase, the obtained classifier model is tested in binary-class SVM with Data 2 which has never been learned by the model. The best performance measurement (accuracy in our binary-class test phase) achieved is stored as the final successful classification result.

As mentioned above, to avoid the variations due to the random selection of data, we conducted the experiment 3 times and took the average of the 3 obtained accuracy values for each IDUL as the final performance measurements.

5 | RESULTS

The collection of positive data took 1 Canadian semester (15 weeks). After eliminating all invalid data, we finally had 1387 keystroke signatures of 36 students in total. The quantity of keystroke signatures for each student varies from 194 to 12 in the dataset. The collection of negative data took 2 weeks. After removing all invalid data, we finally had a total of 2247 keystroke signatures associated to a total of 36 students.

5.1 | Grid search

However, there is no exact formula or algorithm to set its 2 main hyper-parameters γ and ν for tuning the frontier.

[‡] k-fold cross-validation: The training set is split into k smaller sets. For each of the k "folds," a model is trained using k-1 of the folds as training data, and the resulting model is tested on the remaining part of the data to compute the accuracy. The accuracy reported by k-fold cross-validation is then the average of the values computed in the loop.

[§] Accuracy: $\frac{(TP+TN)}{TP+FP+TN+FN}$, where TP, TN, FP, and FN denote respectively true positive, true negative, false positive, and false negative.

[¶] Recall: $\frac{TP}{(TP+FN)}$ (also called true positive rate) measures the proportion of positives that are correctly identified.

Then, we first used a grid search method for searching the best value pair (γ, ν) for each IDUL with LibSVM. This could be regarded as tuning the frontier of classification by trying out all combinations of γ and ν values in a certain range to see the results. At the end, we chose the value pair (γ, ν) with the best recall measurement achieved during the training phase.

Various pairs of (γ, ν) values are tried and the one with the best cross-validation recall is picked. We found using exponentially growing sequences of γ and ν is a practical way to identify good parameters.

a. γ varies from 2^{-40} to 2^{-13} with steps of 2^{1}
b. ν varies from 2^{-10} to 2^{-6} with steps of $2^{0.1}$
c. in this way, 1080 pairs were tried.

We then searched the tendency of recall obtained for each IDUL in the training phase. As explained previously, we tried 3 random selections in the positive data to divide it into 2 parts before the training phase. We obtained 3 groups of best recall and hyper-parameters value pairs (γ, ν). If we draw all the recall values obtained in the 3 experiments for each IDUL in terms of their signature numbers in descending order, we find that only a few signatures showed wide variation of recall. For example, as shown in Figure 4, wide variation of recall was achieved in the case of IDUL #29. This means that only a few positive signatures could be recognized with a best model which could be obtained in training phase, while a lot of positive signatures were rejected.

Figure 5 shows how the IDULs are ranked.

Then, we checked the performance of the models obtained in the training phase. We used the collected negative data with the 3 selections of positive data. For each selection of positive data, we add the same number of negative data to obtain the Data 2. The performance measurement in the test phase is given by the recognition rate accuracy which varies from 98% to 55% with an average value of 85% and a standard deviation of 11.2. The average accuracy obtained in 3 experiments for each IDUL was sorted in terms of the number of its signatures in descending order. We can see the trend by means of the gray line on the top of the Figure 6. It points out that the accuracy tends to decrease slightly with the number of signatures decreasing (from 197 to 12). However, it appears that the number of signatures has no influence on the false positive number (green trend line at the bottom of the figure). Although, for some signatures, the recall value was good for the model training (see Figure 4), we found that the performance can be poor with low accuracy value (Figure 6). For instance, the IDUL 17 showed quite a high recall value in the training phase, while the obtained accuracy was quite low in the testing phase. This may be principally caused by the high false positive rate (see the green column lines in Figure 6).

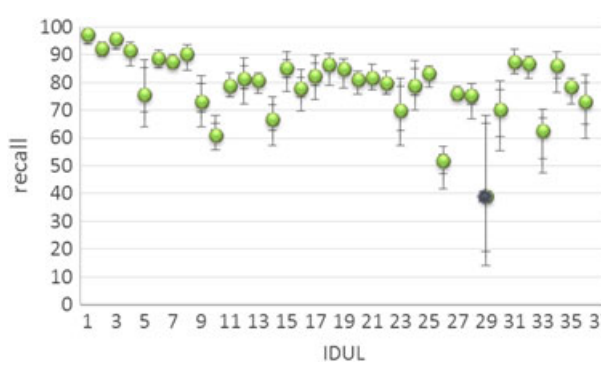


FIGURE 4 Best recall for each IDentifiant Universite Laval (IDUL) when they are sorted in terms of their signature numbers in descending order [Color figure can be viewed at wileyonlinelibrary.com]

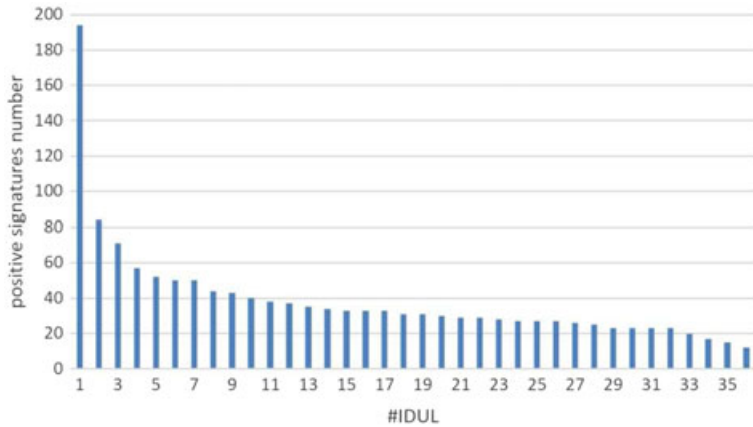


FIGURE 5 The IDULs are ranked in order of positive signature number. IDUL, IDentifiant Universite Laval [Color figure can be viewed at wileyonlinelibrary.com]

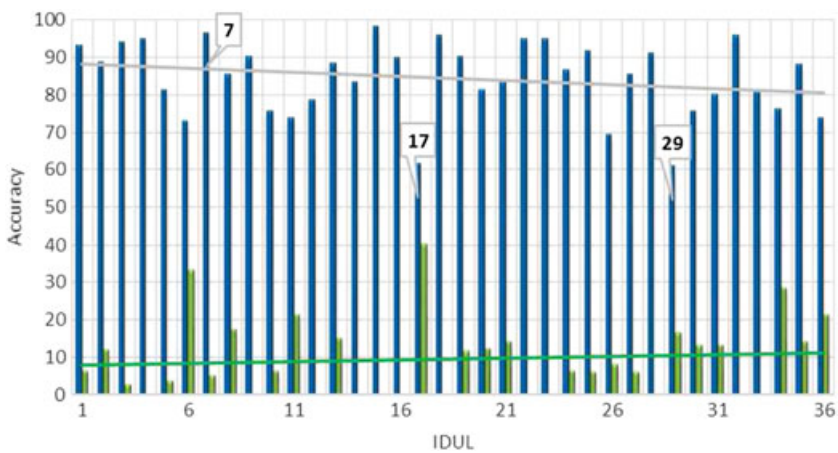


FIGURE 6 Evaluation of the performance by the accuracy measurement. The accuracy tends to decrease slightly with the number of signatures decreasing (gray line). But the number of signatures seems to have no influence on the false positive number (green line). IDUL, IDentifiant Universite Laval [Color figure can be viewed at wileyonlinelibrary.com]

Figure 7 shows the influence of the IDUL length (quantity of characters included in each IDUL) on the performance, by displaying the accuracy as well as false positive rate in terms of IDUL length in descending order. We can observe that the accuracy decreases slightly but the false positive rate increases rapidly with the decrease of IDUL length. Therefore, the IDUL length imposed by the registration system is also an important factor to consider.

The results obtained with the grid search method are generally satisfactory. But, because of its calculation complexity, this method is suitable for only a few parameters.[14] Otherwise, this calculation during the training phase becomes too expensive to find the best parameters.[15] To both reduce the complexity and increase the performances, we are looking for algorithms to outperform the grid search method by making a better choice for the values of the two hyper-parameters of the OCSVM using RBF kernel. It extends our previous works.[16]

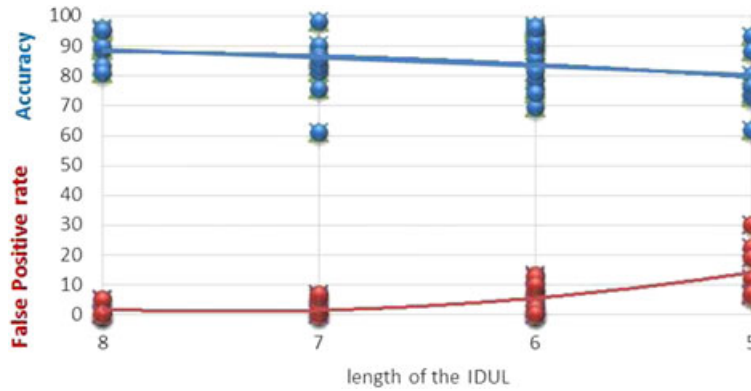


FIGURE 7 Evaluation of the performance: Accuracy decreases slightly but the false positive rate increases rapidly with the decrease of IDentifiant Universite Laval (IDUL) length [Color figure can be viewed at wileyonlinelibrary.com]

5.2 | Indirect method: DFN

We first retained the works of Xiao et al[17] for an indirect method: DFN[#]. This method consists in optimization of the RBF kernel parameter s(which is $1/\gamma$) using the data of only one-class.

This independent OCSVM model method solves the problem by considering information on the longer distances between training samples to their farthest neighbors and their nearest neighbors. Since there is no negative class sample, the samples located on the border are used instead of the negative class samples. They are called pseudo-negative samples. The pseudo-negative samples which reflect local structure can describe the distribution of all samples. In addition, based on the monotonicity of the exponential function of the RBF kernel, a distance inequality in the original space maintains the same relation in the features space after the projection[‖]. This projection by the RBF kernel maintains the same spatial structure as that for these from training samples in the original space.[17] Therefore, if the OCSVM model boundary can separate positive samples from pseudo-negative samples in the case of one-class, then it can also separate positive samples from true negative samples in the case of 2 classes. Xiao et al[17] define the following objective function which should be maximized to get the optimal s:

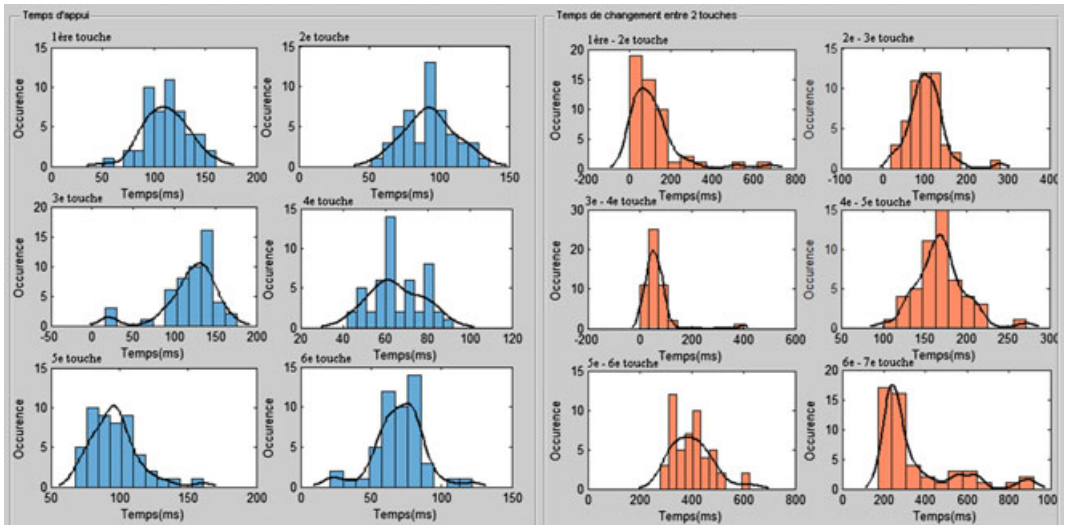
$$f(s) = \frac{1}{n}\sum_{i=1}^{n} \max_{j} \|\phi(\mathbf{x}_i) - \phi(\mathbf{x}_j)\|^2 - \frac{1}{n}\sum_{i=1}^{n} \min_{j\neq i} \|\phi(\mathbf{x}_i) - \phi(\mathbf{x}_j)\|^2 \tag{4}$$

The first term corresponds to the mean of the distance between the farthest samples in the features space. It is a measure of the distance between positive and pseudo-negative samples. The second term corresponds to the mean of the distance between the nearest neighbors. It is a measure of local structure representation quality.[17] Using the monotonicity of the exponential function, Xiao et al[17] propose to simplify this objective function as

$$f(s) = \frac{2}{n}\sum_{i=1}^{n} \exp\left(-\frac{Near(\mathbf{x}_i)}{s}\right) - \frac{2}{n}\sum_{i=1}^{n} \exp\left(-\frac{Far(\mathbf{x}_i)}{s}\right) \tag{5}$$

$$where \quad Near(\mathbf{x}_i) = \min_{j\neq i} \|\mathbf{x}_i - \mathbf{x}_j\|^2 \text{ and } Far(\mathbf{x}_i) = \max_{j} \|\mathbf{x}_i - \mathbf{x}_j\|^2$$

[#] distance to farthest neighbor.
[‖] If $|x_i - x_j| < |x_l - x_k|$ in the original space, then we have $|\phi(x_i) - \phi(x_j)| < |\phi(x_l) - \phi(x_k)|$ in the features space.



(A) Dwell time (B) Flight time

FIGURE 8 Keystroke time features which have monomodal and concentrated distribution (keys 1 to 6) [Color figure can be viewed at wileyonlinelibrary.com]

For a given sample, these two values $Near(\mathbf{x}_i)$ and $Far(\mathbf{x}_i)$ are constant for any x_i. Briefly, the distance between each sample pair should be calculated and the distance between nearest and farthest samples should be found. The local structure maximizing the distance between positive and potential negative class is taken into account. But Xiao et al[17] have shown that it is not possible to obtain satisfactory results if the DTL method is used with multimodal distribution datasets. In fact, the farthest and the nearest distances between samples do not provide sufficient information to characterize the subsequent distribution, particularly for multimode datasets.

So, we did a preliminary statistical study on the signatures before assessing the DTL performance with our data. While most of the time feature distributions are monomodal and concentrated, as shown in Figure 8, multimodal and dispersed distributions are unfortunately often found for some features (for example, the dwell time 1, 3, 5, and 6 in Figure 9)

This preliminary statistical study shows that the DFN method unfortunately cannot be used with our dataset because of the presence of multimodal and dispersed distribution of certain temporal characteristics.[17] We initiated a move toward another method to address the multimodal distribution problem.

5.3 | Direct method: DTL

We retain the direct method DTL** also proposed by Xiao et al.[17] It combines the optimization of the RBF kernel parameter s (which is $1/\gamma$) using the feedbacks from OCSVM models. The basic idea is to define an appropriate decision boundary which is neither too tight to keep the classifier generalization nor too loose to ensure the outliers sensibility. The calculation of samples distances is then used. If there is no training sample inside the decision boundaries, they are judged to be loose. If the boundaries near 2 neighboring samples are concave or disconnected, they are judged to be tight. [17,18]

The DTL method detects the OCSVM model tightness by training samples locations relative to decision boundaries. In comparison, DTL outperforms DFN for the RBF kernel parameter s selection

**detection of tightness or looseness.

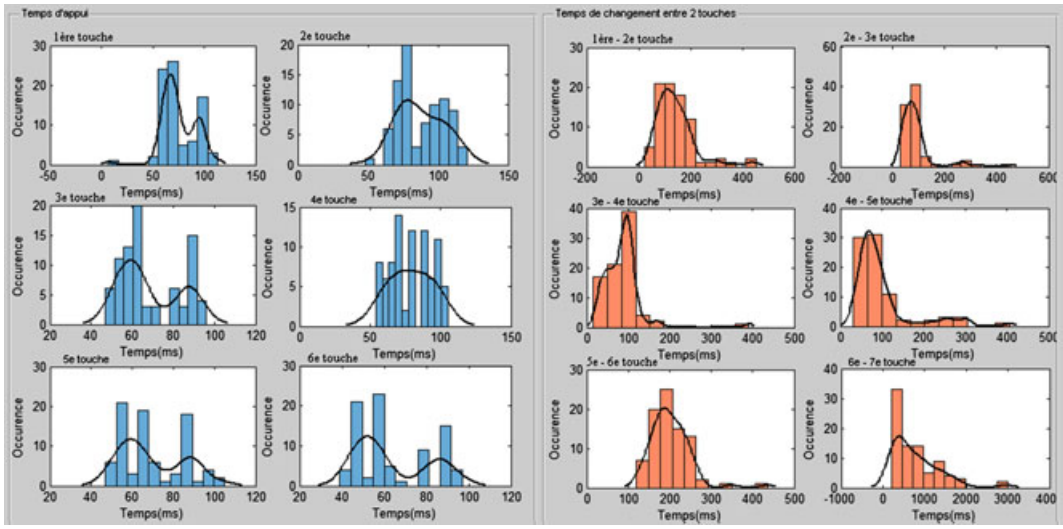


FIGURE 9 Keystroke time features which have multimodal and dispersed distribution (keys 1 to 6) [Color figure can be viewed at wileyonlinelibrary.com]

when samples may have a Gaussian multimodal distribution. According to Xiao et al,[17] the feedbacks from OCSVM models provides useful information from training samples to optimize the parameter s.

However, DTL consists solely of optimizing s, considering ν with an a priori fixed value. As explained previously, ν tunes the upper bound on the fraction of outliers outside the decision boundaries and the lower bound on the fraction of support vectors in the training dataset. The values of s found therefore corresponds to a local optimal value for a (s, ν) combination. The parameter ν must be considered to find the best model. Then, we propose an extended method to search not only a better s but also a better value pair (s, ν).

5.4 | Extended direct method: DTL in greedy search

The aim is to use a greedy method to find the best value pair (γ, ν). We repeat the DTL process with different values of ν in a predefined range (for example, between 0.001 and 0.2). We then search the best value pair(γ, ν) according to the obtained performance score of the OCSVM model. The details of our algorithm are listed in Table 1.

We conduct a series of experiments on our keystroke (IDUL) datasets to test the performance of the proposed method. ν is chosen between 0.001 and 0.02 in a logarithmic scale, and γ is chosen by using the DTL algorithm. Training is processed with only positive data, and the test is processed with positive and negative data. To avoid variations caused by random data selection in the dataset, we repeat the experiment 3 times. The average of the 3 accuracy values obtained for each IDUL is retained as the final performance measure.

The accuracy of the proposed method is compared with those of the grid search method. The results are presented for each IDUL when they are sorted in terms of their signature numbers in descending order (Figure 10). We can then observe that accuracy generally decreases faster with the grid search than with our proposed method when the signature number decreases.

The complete results (grid search compared with our proposed method) are shown in a condensed way in Table 2. Although the average, the median, and the maximum are slightly better with the grid search, it is important to note that the standard deviation is better with our approach (more than 18% lower). It denotes that our method is more robust. Then, we must study the results in more depth.

TABLE 1 Detection of tightness or looseness (DTL) used in greedy search algorithm

	Input: Data1= $(x_1, \ldots, x_n)$**, Data2=** $(z_1, \ldots, z_m)$**,** n_i, n_s, ν **% Training with positive data**
1:	for ν in $[n_i, n_s]$
2:	s ← DTL(ν, Data1)
3:	$\gamma = 1/s$
4:	Recall ← svm_train(γ, ν, v, Data1)
5:	if Recall >BestRecall
6:	BestRecall ← Recall
7:	Bestgamma ←γ
8:	Bestnu ←ν
9:	end_if
10:	end_for
11:	model ← svm_train (Bestgamma, Bestnu, Data1)
12:	accuracy← svm_predict(model, data2) % Test with positive and negative data
13:	return Accuracy
	Output: Accuracy

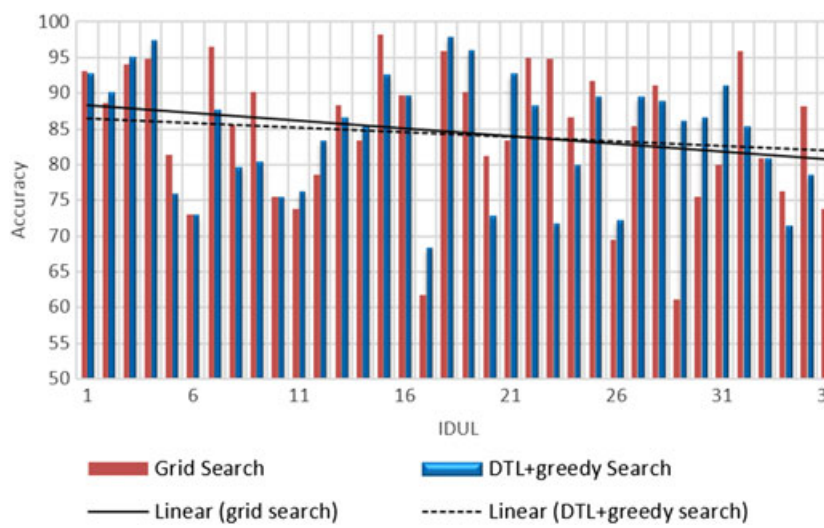


FIGURE 10 Grid search and DTL+greedy search accuracy for the 36 IDULs. DTL, detection of tightness or looseness; IDUL, IDentifiant Universite Laval [Color figure can be viewed at wileyonlinelibrary.com]

Furthermore, if we observe false positive rate for the 2 methods (Figure 11 shows the false positive rate for the grid search and the DTL+greedy search and highlights the lowest rates). We can see that with grid search impostors (false positive case) are more often accepted than with our proposed method (black bar in Figure 11).

Indeed, as shown in Table 3, the DTL+greedy search performance is much better in average, median, standard deviation, and maximum.

These last results highlight the better and more stable performances obtained by using our proposed method even though its calculation complexity is not less expensive. Indeed, better results in false positive rate mean that impostors can be better identified. Then, our proposed method responds well to the aim that we want to reach: verifying the identity of an individual with the constraints of academic systems.

TABLE 2 Average, median, maximum, minimum accuracy, and percentage variation for the 2 methods

Accuracy	Grid search	DTL+ greedy search	Percentage variation grid search vs greedy search
Average (%)	**84.52**	84.21	−0.4
Median (%)	**86.09**	85.76	−0.4
Standard deviation	9.70	**8.21**	**−18.1**
Max (%)	**98.15**	97.92	−0.2
Min (%)	61.11	**68.33**	+10.6

Abbreviation: DTL, detection of tightness or looseness.

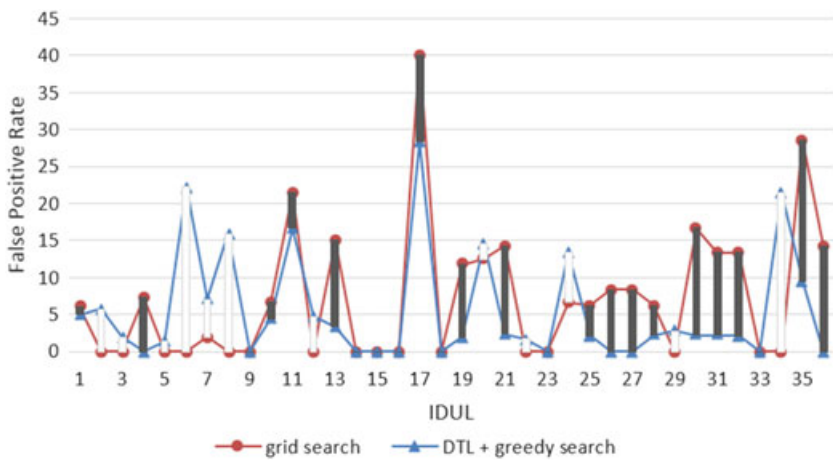


FIGURE 11 Grid search false positive rate for the 36 IDULs. White bars show when false positive rate is higher with the proposed DTL+greedy search method. Black bars show when false positive rate is higher with the grid search method. DTL, detection of tightness or looseness; IDUL, IDentifiant Universite Laval [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 Average, median, maximum, minimum false positive, and percentage variation of the 2 methods

False positive rate	Grid search	DTL+ Greedy search	Percentage variation Grid search vs greedy search
Average (%)	9.40	**8.77**	**−7.2**
Median (%)	6.56	**6.46**	**−1.5**
Standard deviation	10.15	**9.45**	**−7.4**
Max (%)	40	**35**	**−14.3**
Min (%)	0	0	

6 | CONCLUSION

We were interested in student identification for a distance higher education assessment system. This system is based on a unique identifier (IDUL is a short string) given to each student when they enroll the Laval University. The IDUL is regularly and naturally entered by the students. Consequently, this IDUL is used to construct signatures for a biometric keystroke dynamics identification. We then build our own dataset of IDULs with students who followed a remote course at Laval University for 1 semester. Time features are therefore extracted from collected keystroke dynamics and used in a one-class

classification process. Indeed, the identification system needs such a classification because it is not possible to obtain negative data simulating impostors. OCSVM corresponds to such an approach which learns a decision function for anomaly detection. We chose to use RBF kernel, but the main difficulty is finding its better hyper-parameters γ andν which allow the best generalization and then the best accuracy. For that purpose, we retain 3 methods to find a method specially adapted to the targeted application. The first method, called grid search, is a greedy method based on the best recall. The second called DFN is based on the distances between training samples to their neighbors. The third method called DTL uses both feedback of the OCSVM models and the sample neighborhood.

We found that the time features extracted from IDUL and used in the OCSVM process can have multimodal distributions. Therefore, we cannot use the DFN method which is only based on samples characteristics. Furthermore, the DTL method proposed by Xiao et al[17] consists solely of optimizing s (which is $1/\gamma$), considering ν with an a priori fixed value. Then, we proposed to use DTL in a greedy search to find the best value pair (γ ,ν).

We compared the performance of the grid search method and our proposed method. Our experience has shown that the main difficulty for student identification does not necessarily come from the collected signature number. Indeed, we observed that it is more important to collect homogenous signatures. The error rate decreases when the signatures number increases whereby the authentication becomes more robust at each new acquisition. Therefore, the IDUL length has more impact on the false positive rate than the signature number.

After only a few weeks' collection (a dozen signatures is the minimum), we can identify students with an accuracy average of more than 84% but more importantly with a 8.77% false positive rate average, around 7% less than with the grid search even though its calculation complexity is not less expensive. That shows that our proposed method is more robust to identify impostors.

Future work aims at using other kernels than RBF for resolving the problem of multimodal distributions. Combining kernels is one possible way to merge multiple information sources and allow learning different modes separately. Finally, we could add a supplemental factor to the data vector of keystroke dynamics by including the Shift and Caps Lock keys, which are normally used to identify how an individual enters letters in upper case or lower case. It will help to reduce the error rates of the classifier.

ACKNOWLEDGMENT

T. Eude gratefully acknowledges the financial support of Cisco.

ORCID

Thierry Eude http://orcid.org/0000-0002-8129-3460

REFERENCES

1. Banerjee SP, Woodard DL. Biometric authentication and identification using keystroke dynamics: A survey. *J Pattern Recognit Res*. 2012;7(1):116-139.
2. Karnan M, Akila M, Krishnaraj N. Biometric personal authentication using keystroke dynamics: a review. *Appl Soft Comput*. 2011;11(2)1565-1573.
3. Monrose F, Rubin AD. Keystroke dynamics as a biometric for authentication. *Future Gener Comp Sy*. 2000;16(4):351-359.
4. Teh PS, Teoh ABJ, Yue S. A survey of keystroke dynamics biometrics. *Scientific World J*. 2013;2013:1-24.
5. Bleha S, Slivinsky C, Hussien B. Computer-access security systems using keystroke dynamics. *IEEE T Pattern Anal*. 1990;12(12):1217-1222.

6. Robinson JA, Liang VW, Chambers JAM, MacKenzie CL. Computer user verification using login string keystroke dynamics. *IEEE T Syst Man Cy A*. 1998;28(2):236-241.
7. Pisani PH, Lorena AC. A systematic review on keystroke dynamics. *J Brazil Comput Society*. 2013;19(4):573-587.
8. Sang Y, Shen H, Fan P. Novel impostors detection in keystroke dynamics by support vector machine. *In: Parallel and Distributed Computing: Applications and Technologies*. Berlin/Heidelberg: Springer; 2005:666-669.
9. Wei Hsu C, Chung Chang C, Jen Lin C. A practical guide to support vector classification; 2010.
10. Chang C-C, Lin C-J. Libsvm: a library for support vector machines. *ACM Trans Intell Syst Technol*. 2011;2(3):27:1-27:27. ISSN 2157-6904. https://10.1145/1961189.1961199.ISSN
11. Yu E, Cho S. Novelty detection approach for keystroke dynamics identity verification. *In: Intelligent Data Engineering and Automated Learning*. Berlin/Heidelberg: Springer; 2003:1016-1023.
12. Schölkopf B, Platt JC, Shawe-Taylor J, Smola AJ, Williamson RC. Estimating the support of a high-dimensional distribution. *Neural Computation*. 2001;13(7):1443-1471.
13. Joyce R, Gupta G. Identity authentication based on keystroke latencies. *Commun ACM*. 1990;33(2):168-176.
14. Friedrichs F, Igel C. Evolutionary tuning of multiple svm parameters. *Neurocomputing*. 2005;64:107-117.
15. Li J, Zhang C, Li Z. Battlefield target identification based on improved grid-search SVM classifier. *In: 2009 International Conference on Computational Intelligence and Software Engineering*, CISE 2009, December 8-10, 2009; Wuhan, China, IEEE, pp. 1-4.
16. Chang C, Eude T, Obando Carbajal LE. Biometric authentication by keystroke dynamics for remote evaluation with one-class classification. *Advances in Artificial Intelligence: 29th Canadian Conference on Artificial Intelligence*; May 31-June 3, 2016; Victoria, BC, Canada. Cham, Switzerland: Springer International Publishing; 2016:21-32.
17. Xiao Y, Wang H, Zhang L, Xu W. Two methods of selecting gaussian kernel parameters for one-class svm and their application to fault detection. *Knowl-Based Syst*. 2014;59:75-84.
18. Wang H, Zhang L, Xiao Y, Xu W. An approach to choosing gaussian kernel parameter for one-class svms via tightness detecting. *2012 4th International Conference on Intelligent Human-Machine Systems and Cybernetics (IHMSC)*, vol. 2. Jiangxi, China: IEEE; 2012:318-323.

How to cite this article: Eude T, Chang C. One-class SVM for biometric authentication by keystroke dynamics for remote evaluation. *Computational Intelligence*. 2018;34:145–160. https://doi.org/10.1111/coin.12122

Copyright of Computational Intelligence is the property of Wiley-Blackwell and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.